Sensorimotor control of robots mediated by electrophysiological measurements of fungal mycelia

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Abstract

Living tissues are still far from being used as practical components in biohybrid robots because of limitations in life span, sensitivity to environmental factors, and stringent culture procedures. Here, we introduce fungal mycelia as an easy-to-use and robust living component in biohybrid robots. We constructed two biohybrid robots that use the electrophysiological activity of living mycelia to control their artificial actuators. The mycelia sense their environment and issue action potential-like spiking voltages as control signals to the motors and valves of the robots that we designed and built. The paper highlights two key innovations: first, a vibration- and electromagnetic interference-shielded mycelium electrical interface that allows for stable, long-term electrophysiological bioelectric recordings during untethered, mobile operation; second, a control architecture for robots inspired by neural central pattern generators, incorporating rhythmic patterns of positive and negative spikes from the living mycelia. We used these signals to control a walking soft robot as well as a wheeled hard one. We also demonstrated the use of mycelia to respond to environmental cues by using ultraviolet light stimulation to augment the robots' gaits.

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